

Agile Process & Individual Reflection

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Course : DS26 - Data Science and Business

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1. Individual role and contribution

My role in this project was the product analysis section. Essentially, this entails I own everything from the initial data exploration at the product level, all the way to the final findings and recommendations. My deliverables include : teh entire product analysis code in the shared Jupyter notebook; six charts covering several analyses from the Pareto distribution, the price band analysis, the ceramic jar investigation, the September 2010 spike, EIRE recurring cancellations and the key account product breakdown across four countries.

Beyond my core section, I equally contributed to the establishment of the business problem, the redaction of the certain parts of the final report (including but but limited to the methods section). Within the product analysis itself, I took on an additional layer of investigation : a product breakdown for the key accounts identified in the customer analysis to determine whether a product issue was sitting underneath the account level patterns.

Another additional effort worth mentioning is the StockCode M cleaning decision. During my exploration of the dataset, I noticed that manual financial adjustment codes were being counted as product cancellations, significantly inflating the loss figures for some countries (EIRE, Norway and France). Flagging and resolving this matter changed the conclusion for Norway entirely and sharpened the findings for the other markets.

2. Lessons learned

The biggest challenge I faced was acquiring technical findings and translating them into business insights. Early in the project, I had a hard time grasping what the actual business problem was behind the data. I was approaching the entire topic through the lens of code (running analyses, producing outputs, interpreting numbers...) without asking what any of it

actually meant for a real business. My first version of the product analysis reflected that. I had coded the section without fully integrating the geographic and customer findings that came before it, and when I stepped back and read what I had written, the insights weren't holding up. I deleted the whole thing and started over with fresh eyes.

Starting over was the right call. The second version was built around the business questions first and was not just randomly solving unnecessary issues. Making that complete 180 in my way of thinking about the work (from developer to decision maker) was extremely uncomfortable at first but ended up being the most important adjustment I made throughout the project.

Something I would definitely do differently is make my mental shift much sooner. I spent a lot of time producing inconsistent and horrible work which I ultimately discarded because I hadn't clearly defined what I was trying to answer before I started building. This stumble made me more conscious and from now on for my projects, I spend more time at the start mapping the theoretical part explicitly before writing a single line of code.

3. Collaboration process

Our team collaborated through a combination of a group chat for day to day communication, scheduled meetings / calls among the four of us, and a separate Teams channel with our professor where she monitored our progress checkpoint by checkpoint. The checkpoint structure was important; it forced the team to establish a workable pipeline, deliver incrementally rather than trying to to assemble all of it at the end, and it gave the professor visibility into how the work was distributed.

The team leader played a central coordination role. He integrated the three workstreams, helped me out with insightful input on the quality and usefulness of my work. Communication within the team was mostly asynchronous between meetings, with the group chat used to share updates, deliverables, flag blockers and align on framing before each checkpoint submission.

4. Learning outcomes

- a. **Technical** : the most significant technical learning was understanding how to structure an end-to-end analysis; from business problem definition through to EDA, findings, interpretation, and recommendations. That full arc is not something I had done before in a way that felt genuinely coherent. I also deepened my practical Python skills working with a large, messy real world dataset, particularly around cleaning decisions, and building charts that communicate findings instead of just displaying data. I also learned to appreciate working on a retail dataset, which was something I previously found quite tedious.
- b. **Business understanding** : before this project, I saw data analysis only as a technical exercise; now I can appreciate it as a communication exercise where the technical work is the means, not the end. Learning to ask questions like "what exactly does this mean for the business?" before "what do these numbers say?" was a genuine shift in how I approach analytical problems now. The product section specifically taught me that the same figure can mean completely different things

based on the context; also that emerging the right context is as much a part of the analyst's job as running the code.

- c. **Project management** : working to a checkpoint based structure with a fixed deadline taught me the value of scoping work clearly before starting. The time I spent rebuilding my section from scratch was a direct consequence of not having done that upfront. I also learned how to work in parallel with teammates who are handling adjacent sections; because being aware of what the geographic and customer analyses were finding helped me shape my own framing. It made the final product all the more coherent. Through this project believe it or not I also learned the importance of time management.
- d. **Analytical thinking** : the Pareto framing that runs through this project is something I will carry forward into future works. More broadly, this project reinforced that good analysis isn't just finding the most interesting pattern; it's also finding the pattern that's actually true, being honest about what it does, and bringing out the insights about what it means for someone who makes decisions.

On a personal note, I would like to add that honestly, having a professor who takes this work seriously and holds high standards was a significant motivation factor. It pushed me to care about the quality and rigour of what I submitted in a way that made the final result something I am genuinely proud of. So, thank you professor.